

ing AI Ability to Complete Long Software Tasks: Time Horizon Doubling Every 7

Headline finding

50% task-completion time horizon has doubled every ~7 months (≈ 207 days) since 2019.

From GPT-2 (~2 seconds) to o3 (~1 hour 50 minutes) in ~6 years.

Authors: Thomas Kwa, Ben West, Joel Becker, et al. (METR)

A New Metric: 50% Task-Completion Time Horizon

Method workflow

- 1) Build a diverse suite of 170 software/research tasks.
- 2) Measure skilled-human completion times and AI success rates on the same tasks.
- 3) Fit a logistic success model and read off the task duration where success probability = 50%.

Interpretation: the “50% horizon” is an intuitive, human-grounded autonomy metric.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Task Suite Source	Task Count	Human-Time Range	Representative Examples
HCAST	97	~1 min to 30 hrs	wikipedia_research (~1 min), munge_data (~56 min)
RE-Bench	7	~8 hrs each	cuda_backtesting (~8 hrs)
SWAA	66	~1 sec to 30 sec	find_shell_script (~3 sec)

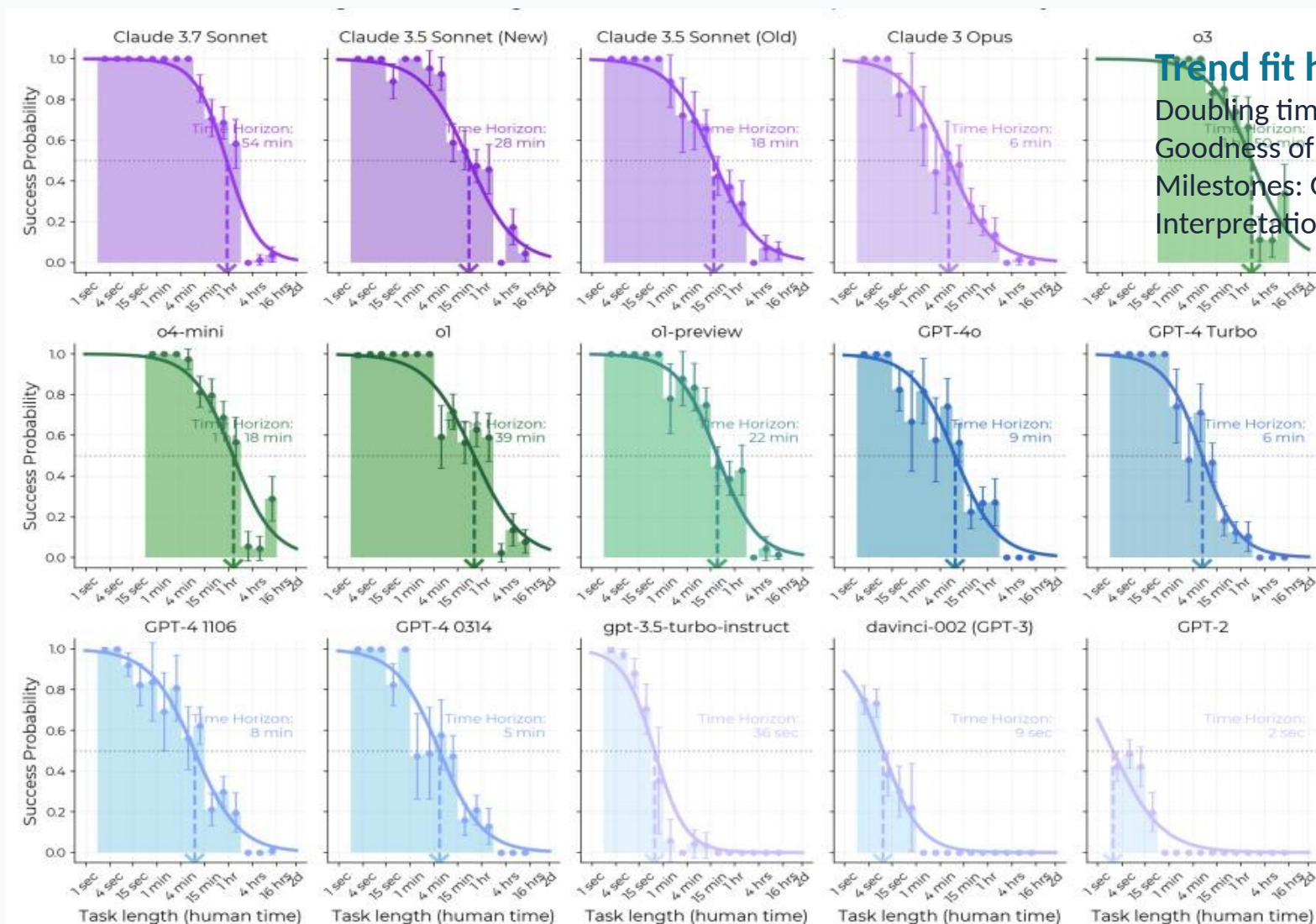
Suite composition and span

Combined benchmark: 170 tasks with >800 human baselines totaling 2,529 hours.

Example duration span represented in this deck: 3 seconds → 8 hours.

Design intent: mix atomic actions, practical software tasks, and hard research-engineering work.

Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years



Trend fit highlights

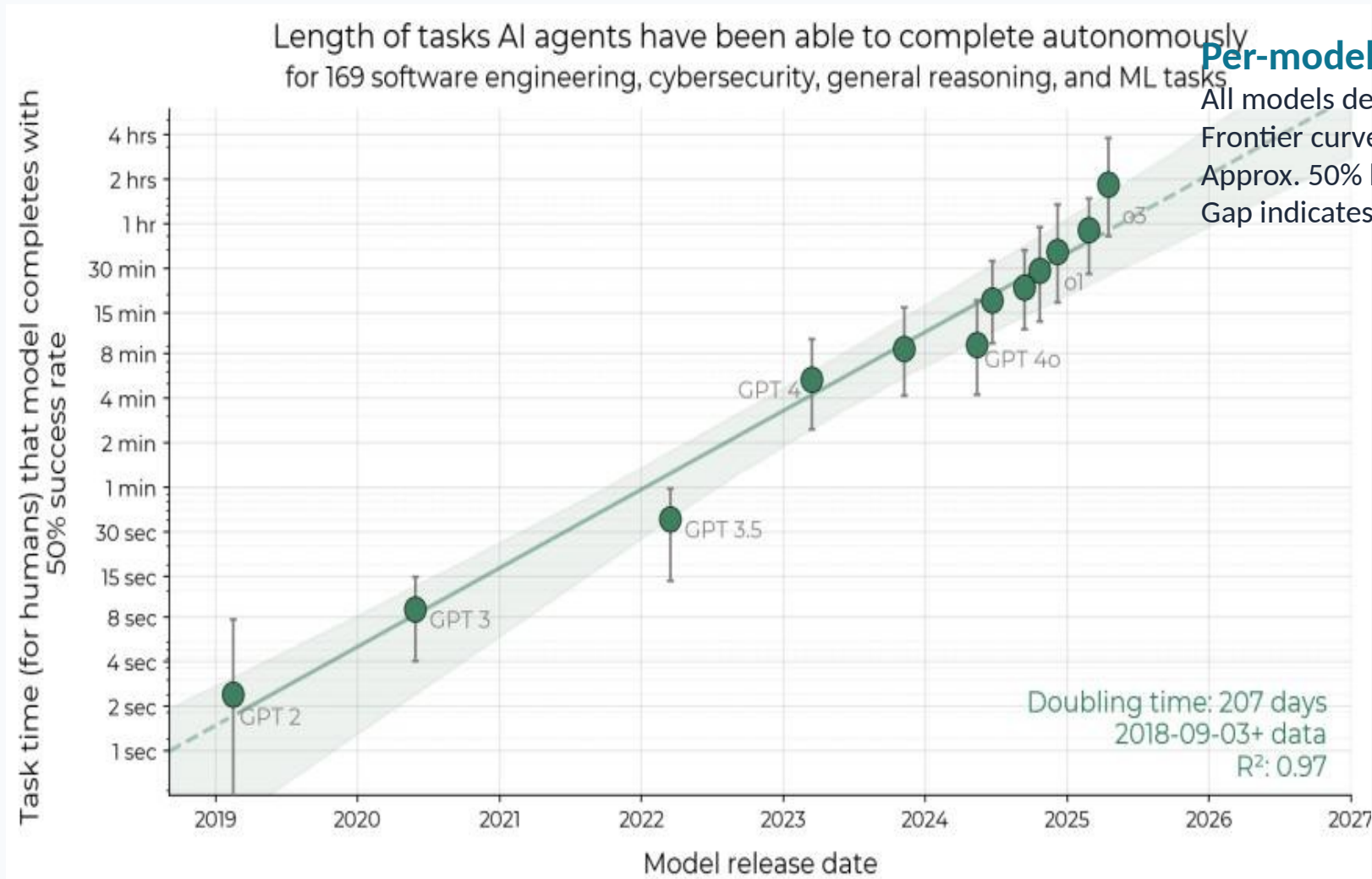
Doubling time: ~207 days (~7 months).

Goodness of fit: $R^2 = 0.97$.

Milestones: GPT-2 (~2s), GPT-4 0314 (~5m), o3 (~1h50m).

Interpretation: sustained exponential growth in autonomous software

Success Probability Drops Sharply as Task Length Increases for All Models



Per-model curve readout

All models degrade with longer tasks.

Frontier curves are shifted right versus older models.

Approx. 50% horizons: GPT-2 (~2s), Claude 3.7 Sonnet (~54m), o3 (~1.5h)

Gap indicates sizable remaining reliability limits on multi-hour tasks

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Observed capability drivers

Improved logical reasoning supports longer dependency chains and planning depth.

Stronger tool use (coding environments, search, debugging) increases execution leverage.

Higher reliability and self-awareness reduce derailment during long trajectories.

Better adaptation to mistakes enables iterative recovery instead of hard failure.

Known limitation

Performance remains materially lower on less structured, “messier” tasks; external validity is still an open question.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Forward projection from historical trend

Naive extrapolation implies crossing a 1-month horizon (~167 work hours) between mid-2028 and mid-2031.

Recent period (2023–2025) shows ~20% faster horizon growth than the 2019–2025 fit.

If trend persists, autonomous systems may absorb larger software projects currently requiring multi-week effort.

Caveats

Trend may bend due to model, data, or deployment constraints.

Benchmark-to-real-world transfer remains uncertain and task-distribution dependent.

ys: Exponential Autonomous Capability Growth Demands Proactive Safety Go

Key takeaways for safety and policy

- 1) 50% time horizon is a robust, intuitive autonomy metric ($R^2=0.97$ across 6 years).
- 2) Capability doubled every ~7 months, with signs of acceleration since 2024.
- 3) Current frontier reaches ~2 hours of skilled-human-equivalent software work (o3).
- 4) External validity is unresolved: benchmark tasks may not fully represent real-world knowledge work.
- 5) Governance focus: monitor horizon growth, stress-test on messy tasks, and align safety controls with rapidly increasing autonomous execution ler